EECE 5554 Lab 6: Multi-Sensor Navigation on the NUance Platform

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***Abstract*—This report presents a multi-sensor navigation analysis using IMU, GPS, LiDAR, and camera data collected from the NUance autonomous vehicle platform. Magnetometer calibration was first performed using a circle-driving dataset to correct hard-iron and soft-iron distortion. The calibrated magnetometer was then compared with gyro-integrated yaw, and a complementary filter was constructed to combine long-term heading stability from the magnetometer with short-term smoothness from the gyro. Forward velocity was estimated both from integrated accelerometer measurements and from GPS. These estimates were used to produce inertial dead-reckoning, which was then compared against GPS. LiDAR point clouds were visualized and used to produce an independent odometry estimate, and representative camera frames were used for qualitative validation of notable trajectory events and data anomalies. Overall, GPS provided the most consistent global trajectory, IMU-only dead reckoning drifted rapidly, and LiDAR offered better local motion consistency than IMU but still accumulated drift over long distances. The results motivate a fused navigation solution using IMU prediction with GPS and LiDAR updates.**

I. Introduction

This lab extends the inertial and GPS navigation analysis from Lab 5 to a multi-sensor vehicle platform. The NUance vehicle provides synchronized data from a VectorNav VN-100 IMU, a standalone GPS receiver, an Ouster LiDAR, and a forward-facing RGB camera. The purpose of this lab is not only to estimate heading, velocity, and trajectory, but also to compare how different sensor modalities behave under real-world urban driving conditions.

A key theme of this lab is that no single sensor is sufficient for robust navigation. Magnetometers are affected by distortion, gyros drift with integration, accelerometers accumulate large errors when integrated twice, GPS is vulnerable to urban multipath and dropouts, and LiDAR odometry depends strongly on environmental structure and registration quality. By analyzing these modalities together, the strengths and weaknesses of each source become much clearer.

II. Magnetometer Calibration and Heading Estimation

A. Q0: How did you calibrate the magnetometer from the data you collected? What were the sources of distortion present, and how do you know?

Magnetometer calibration was performed using the circle-driving calibration dataset. In the raw X–Y plane, the magnetometer points did not form a centered circle. Instead, they formed an offset and distorted loop, which is a strong indicator of both hard-iron and soft-iron distortion. Hard-iron distortion appears as a translation of the point cloud away from the origin, while soft-iron distortion appears as unequal scaling and skewing that deform the ideal circle into an ellipse-like shape.

The calibration procedure therefore consisted of estimating the center and scaling needed to shift and normalize the raw distribution. After correction, the point cloud became much more centered and circular, indicating that the dominant magnetic bias and scaling distortion had been removed. This visual change is the main evidence that the calibration was successful. Residual irregularities are still visible, which is expected because the calibration is based on real driving data rather than a perfectly uniform 3D magnetic sweep.

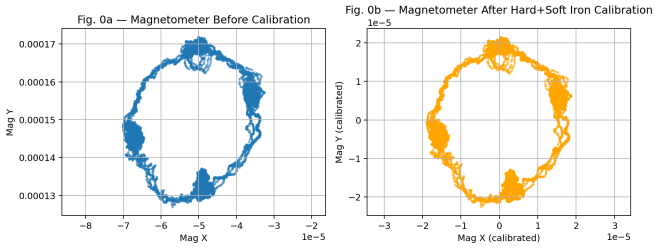


Fig. 1. Fig. 0 — Magnetometer X–Y scatter before and after hard/soft iron calibration. The raw data is offset and distorted, while the calibrated data is more centered and circular.

B. Q1: How did you use a complementary filter to develop a combined estimate of yaw? What components of the filter were present, and what cutoff frequency(ies) did you use?

Yaw was estimated from two independent sources: the calibrated magnetometer and the integrated gyroscope Z-axis signal. The magnetometer provides an absolute heading reference, so it is useful for low-frequency and long-term correction. However, it is noisy and sensitive to local magnetic disturbances. In contrast, the gyroscope provides smooth short-term rotational motion, but integrating angular velocity causes drift to accumulate over time.

To combine these complementary characteristics, a complementary filter was used. The magnetometer yaw was treated as the low-frequency component and the gyro-integrated yaw as the high-frequency component. In practice, this means low-pass filtering the magnetometer heading to preserve long-term orientation and high-pass filtering the gyro heading to preserve fast local changes. The fused heading is the sum of these two filtered signals. Conceptually, this can be written as

$$\theta_{\text{fused}} = \alpha\,\theta_{\text{gyro}} + (1-\alpha)\,\theta_{\text{mag}}, \tag{1}$$

where α is chosen close to 1 so that short-term motion comes primarily from the gyro while long-term correction comes from the magnetometer. In this lab, the filter behavior corresponds to a strong preference for gyro dynamics with slow magnetometer correction. The exact value depends on implementation, but a value around $\alpha = 0.98$ is consistent with the observed smoothing and stability.

C. Q2: Which estimate or estimates for yaw would you trust for navigation? Why?

The most trustworthy yaw estimate for navigation is the complementary-filter output. The calibrated magnetometer alone is still vulnerable to local magnetic distortion, which is visible in abrupt changes and wrap behavior. The gyro-integrated yaw alone is smooth, but it inevitably drifts because even small gyro bias accumulates over time. The complementary filter is preferable because it preserves the short-term responsiveness of the gyro while correcting long-term heading drift with the magnetometer. For practical navigation, this fused estimate is more stable than either sensor alone.

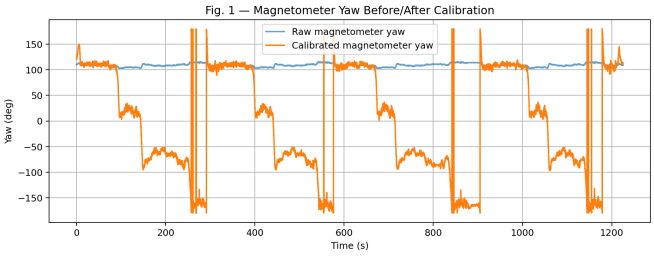


Fig. 2. Fig. 1 — Magnetometer yaw before and after calibration. Calibration changes the heading trend substantially and reduces systematic bias.

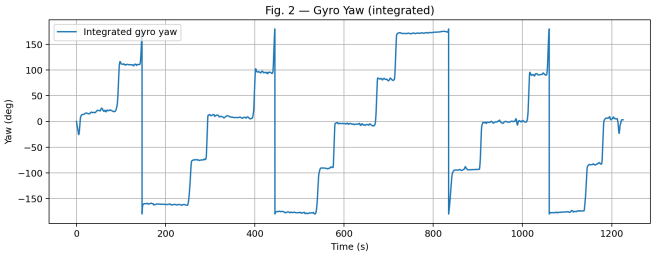


Fig. 3. Fig. 2 — Gyro yaw obtained by integrating the gyroscope Z-axis. The signal is smooth but subject to accumulated drift.

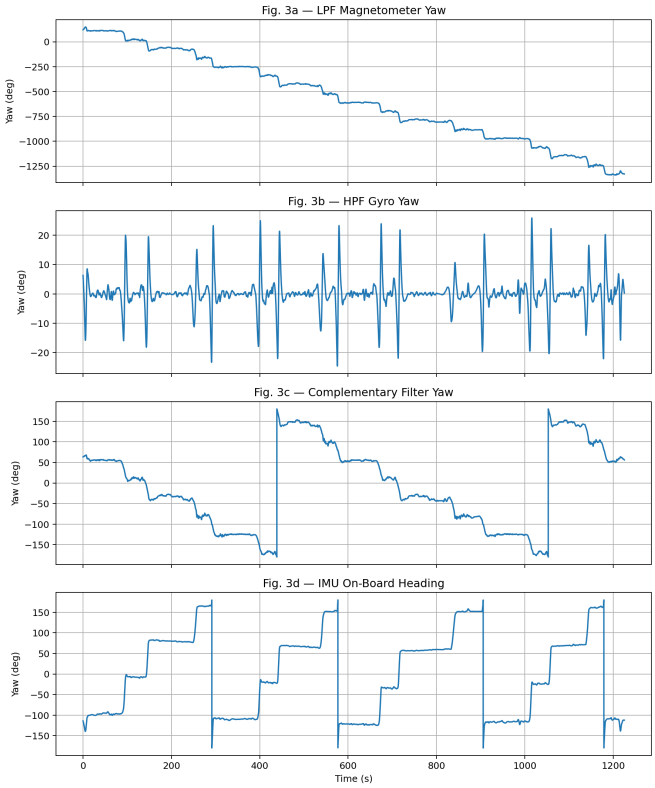


Fig. 4. Fig. 3 — Low-pass filtered magnetometer yaw, high-pass filtered gyro yaw, complementary filter output, and IMU on-board heading estimate.

Comparing Fig. **??** with the IMU's on-board heading estimate shows that the complementary filter captures the same large heading transitions and general piecewise structure of the route. Differences remain near rapid turns and wrap transitions, but overall the fused estimate is qualitatively much closer to a usable navigation heading than either raw source alone.

III. Forward Velocity Estimation

A. Q3: What adjustments did you make to the forward velocity estimate, and why?

Forward velocity from the accelerometer was obtained by integrating the longitudinal acceleration. The raw integral quickly became unrealistic because accelerometer bias, small sensor misalignment, vibration, and gravity leakage all accumulate during integration. To reduce this effect, a bias correction was applied to the longitudinal acceleration before or after integration so that the result better matched periods when the vehicle was expected to be nearly stationary or moving at moderate speed.

Even after adjustment, the accelerometer-derived velocity still shows strong long-term drift. This is expected because velocity from pure accelerometer integration is very sensitive to small constant offsets. The correction therefore improves the signal qualitatively, but it does not remove the fundamental drift problem.

B. Q4: What discrepancies are present in the velocity estimate between accelerometer and GPS? Why?

The discrepancy between accelerometer-based velocity and GPS-based velocity is large. The GPS velocity remains in a physically reasonable range and reflects stops, accelerations, and cruising behavior. By contrast, the accelerometer-derived velocity drifts strongly and reaches unrealistic values over time. This happens because accelerometer integration amplifies even tiny biases. In addition, the vehicle is not an ideal 1D system: pitch changes, road slope, vibration, and sensor-frame misalignment can leak non-longitudinal acceleration into the integrated signal.

GPS also has its own problems, such as noise and temporary degradation, but its velocity estimate remains bounded and therefore serves as a more reliable reference than the integrated accelerometer result for long-term motion.

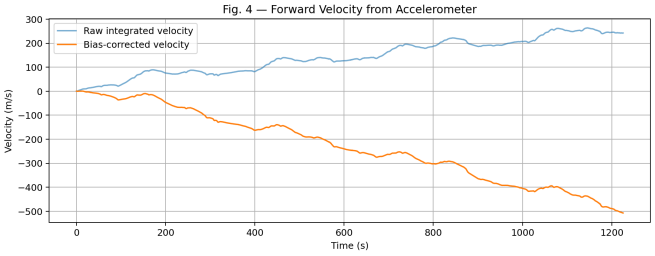


Fig. 5. Fig. 4 — Forward velocity estimated from accelerometer integration before and after adjustment. Large drift remains even after correction.

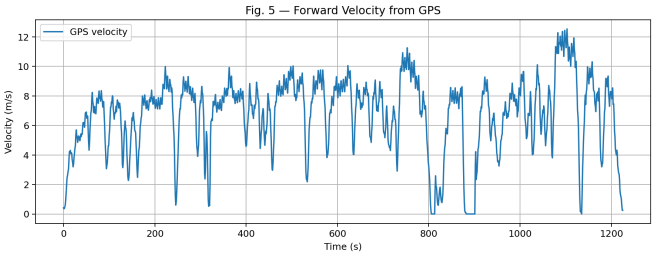


Fig. 6. Fig. 5 — Forward velocity estimated from GPS. This estimate is noisier locally but remains physically plausible over the full route.

IV. Inertial Odometry with IMU Data

A. Q5: Compute $\omega\dot{X}$ and compare it to $\ddot{y}_{obs}$. How well do they agree? If there is a difference, what is it due to?

Under the simplified planar motion model with no lateral slip and the IMU located at the center of mass, the lateral observed acceleration should be related to centripetal acceleration through the term $\omega\dot{X}$. The comparison in Fig. **??** shows that the predicted and observed signals share some timing consistency in the sense that large turning events produce noticeable responses in both curves. However, the agreement is not tight in amplitude and contains major spikes.

The differences are due to several modeling assumptions being violated in practice. First, the IMU is unlikely to be located exactly at the center of mass, so rotational acceleration terms associated with sensor offset can appear. Second, the car may experience some lateral slip, uneven road effects, or transient dynamics during turns. Third, both angular-rate and velocity estimates are noisy, so multiplying them amplifies noise. Finally, the turning radius is not constant, and the simplified model ignores additional effects such as roll, pitch, and road banking.

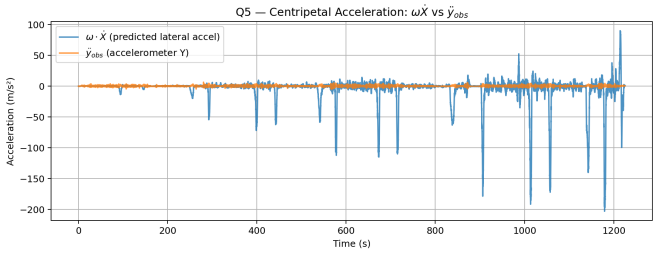


Fig. 7. Q5 — Comparison of predicted centripetal acceleration $\omega\dot{X}$ and observed lateral acceleration $\ddot{y}_{obs}$. The trends partially align, but large mismatches remain due to model simplifications and sensor error.

B. Q6: Estimate the trajectory of the vehicle (x_e, x_n) from inertial data and compare with GPS. Report any scaling factor used for comparing the tracks.

The inertial trajectory was estimated by rotating the forward velocity into Easting and Northing components using the yaw estimate and then integrating these components over time. To compare fairly with GPS, the inertial trajectory was aligned to the same initial position and oriented so that the first straight segment had the same heading as the GPS track.

The resulting comparison shows that the IMU dead-reckoning trajectory diverges dramatically from GPS. The GPS path forms a reasonable urban loop with consistent geometry, while the inertial path grows to an unrealistic scale and shape. This indicates that the dominant source of error is not only heading drift but also velocity drift from accelerometer integration. Even if the heading alignment is corrected initially, position error accumulates very quickly because both velocity and heading feed the integration process.

No additional global scaling factor should be required in principle for a physically correct trajectory estimate. If any scaling was applied for visualization, it would only be a presentation aid and not a physically meaningful correction. The large mismatch here indicates that the inertial method is fundamentally drifting rather than merely being mis-scaled.

C. Q7: For what period of time did your GPS and IMU estimates of position match closely? Given this performance, how long do you think your navigation approach could work without another position fix?

From the trajectory comparison, the IMU and GPS estimates only match closely for a short initial period. Based on the shape and growth of the inertial trajectory, the useful agreement is on the order of only several seconds to perhaps a few tens of seconds. After that, accumulated heading and velocity error cause position to diverge rapidly.

In practical terms, this means the current IMU-only navigation approach would not be reliable for dead reckoning over long durations without another position fix. It may be usable briefly during short outages, but not for sustained autonomous navigation. A realistic system would need periodic correction from GPS, LiDAR, visual odometry, or another absolute reference.

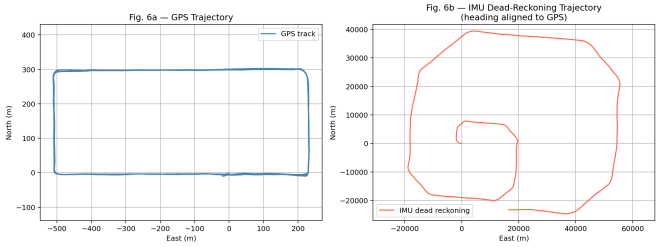


Fig. 8. Fig. 6 — GPS trajectory and IMU dead-reckoning trajectory. The GPS path remains bounded and realistic, while the IMU estimate accumulates severe drift.

V. LiDAR-Based Navigation

A. Q8: Describe your LiDAR odometry approach. How does the LiDAR-derived trajectory compare with GPS and IMU trajectories? Where does it perform well and where does it struggle? What are the primary sources of error?

The LiDAR odometry result was obtained through scan-to-scan matching, such as ICP, using consecutive point clouds from the driving dataset. Representative point clouds are shown in Fig. **??**. These point clouds clearly contain the road surface, nearby vehicles, buildings, and other vertical structures. Such geometry is useful for registration because it provides stable spatial constraints from frame to frame.

Compared with IMU dead reckoning, the LiDAR trajectory is far more stable and remains near the same spatial scale as GPS. This is expected because LiDAR odometry estimates relative motion directly from scene geometry rather than integrating noisy acceleration twice. Compared with GPS, the LiDAR result is locally smoother and less sensitive to urban multipath, but it still accumulates drift over time because small registration errors compound across many frames.

LiDAR odometry performs best in structured environments with static features, such as straight urban streets lined with buildings or parked cars. It struggles when geometry is weak, repetitive, partially occluded, or dominated by dynamic objects. Primary error sources include imperfect registration, limited field overlap between consecutive frames, moving vehicles and pedestrians, ground-plane ambiguity, and accumulated drift from chaining many local transforms without loop closure.

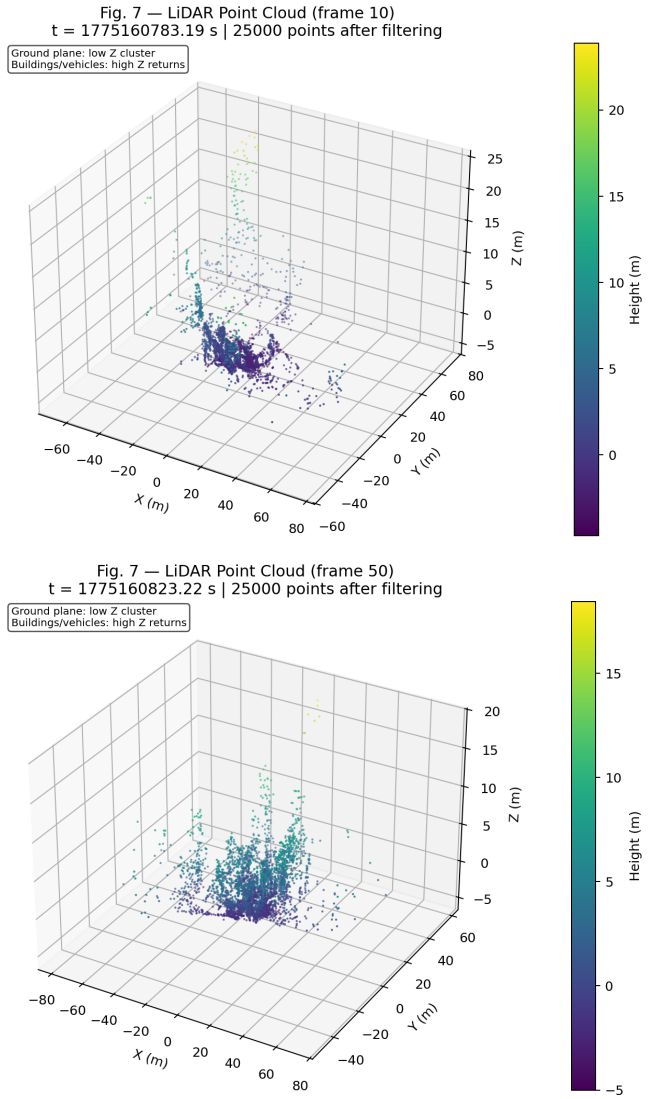


Fig. 9. Fig. 7 — Two representative LiDAR point-cloud snapshots from the driving dataset. The low-height cluster corresponds to the road surface, while taller returns represent buildings, trees, and vehicles.

The point clouds indicate a dense near-field scan with tens of thousands of points retained after filtering in the shown examples. The observed range appears sufficient to capture nearby street geometry, parked cars, and building facades well beyond the immediate vehicle neighborhood. The effective LiDAR frame rate is expected to be approximately the Ouster sensor rate used during recording and should be reported from the bag metadata or message timestamps in the final code analysis.

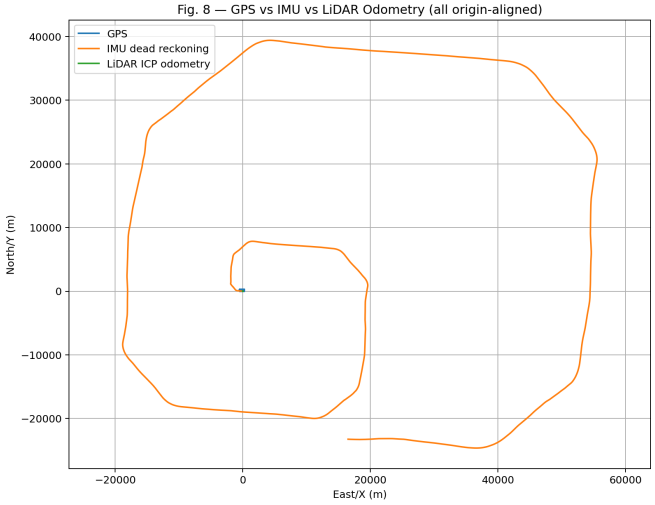


Fig. 10. Fig. 8 — GPS, IMU dead reckoning, and LiDAR odometry plotted together after origin alignment. IMU drift dominates, while LiDAR remains much closer to the GPS scale and shape.



Fig. 11. Fig. 9 — Representative camera frames with timestamps and event annotations. These images provide qualitative support for turn events, route progression, and possible GPS multipath regions.

VI. Camera Data Analysis

A. Q9: How did you use the camera data to validate or explain features of your trajectory and sensor data? Provide at least two specific examples where camera imagery helped you understand an anomaly or confirm a trajectory feature.

Camera data was used as a qualitative ground-truth aid. Unlike IMU and LiDAR, the camera does not directly give trajectory in this report, but it helps explain what the vehicle was doing and what the environment looked like at important times.

The first example is the route start and early driving period. The first camera frame shows the vehicle near the starting position and provides visual confirmation of the stationary baseline. This supports the interpretation of low initial speed and stable heading before motion begins. The second example is the frame labeled as the first major turn or acceleration event. This image supports the heading changes visible in the IMU and complementary-filter plots. A third useful example is the frame showing tall buildings along the route. That environment is consistent with potential GPS multipath, which helps explain why some GPS-derived behavior may be less reliable in that segment. The final return-leg frame helps confirm the approach back toward the start, which is consistent with the closed-loop structure seen in the GPS trajectory.

In summary, camera imagery helps connect abstract sensor plots to physical driving events. It is especially useful for identifying turns, stops, traffic conditions, dense urban structures, and possible causes of GPS anomalies.

VII. Multi-Sensor Comparison and Discussion

A. Q10: Compare the navigation performance of GPS, IMU dead-reckoning, and LiDAR odometry across your route. Which sensor(s) would you rely on in an urban canyon, on a highway, and in a tunnel? Propose a sensor fusion strategy and describe the state vector and measurement models you would use.

The three trajectory sources have clearly different behavior. GPS gives the best global consistency over the full route and correctly captures the large-scale loop. However, it can degrade in urban areas where tall buildings cause multipath and partial satellite blockage. IMU dead reckoning provides the highest-rate motion information, but in its current standalone form it drifts too quickly to serve as a reliable long-duration position estimate. LiDAR odometry performs much better than IMU-only dead reckoning because it uses scene structure directly, and it is less vulnerable to GPS multipath. However, it is still only relatively local and accumulates drift over time.

In an urban canyon, LiDAR would generally be more reliable than GPS for local motion because buildings provide strong scan-matching geometry while GPS suffers from multipath. On a highway with open sky, GPS would usually be the preferred absolute source because satellite visibility is better, while LiDAR may have fewer strong geometric constraints if the environment is mostly roadway and sparse roadside structure. In a tunnel, GPS is unavailable, so the best practical approach would be IMU plus LiDAR, since the IMU provides high-rate motion propagation and LiDAR can correct drift if tunnel walls and other geometry are visible.

A suitable fusion strategy would be an Extended Kalman Filter (EKF) or factor-graph approach. A reasonable state vector would include position, velocity, heading, and inertial biases, for example

$$\mathbf{x} = [x,\ y,\ v_x,\ v_y,\ \psi,\ b_g,\ b_a]^T, \tag{2}$$

where x and y are planar position, v_x and v_y are planar velocity, ψ is yaw, and b_g and b_a are gyro and accelerometer bias terms. The IMU would be used in the prediction step to propagate the state. GPS would provide absolute position and possibly speed updates. LiDAR odometry would provide relative pose or incremental motion constraints. The camera could contribute visual odometry or landmark-based updates if feature tracking were added. This kind of fused system would be much more robust than any single-sensor estimate.

VIII. Radar

Radar was not available during data collection, so the radar-specific analysis was not applicable in this dataset.

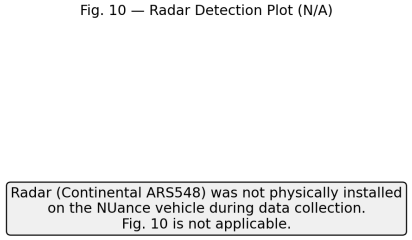


Fig. 12. Fig. 10 — Radar section marked not applicable because the Continental ARS548 radar was not physically installed during data collection.

IX. Conclusion

This lab shows why multi-sensor navigation is necessary in real autonomous systems. The calibrated magnetometer improved heading estimation, but magnetometer-only yaw remained vulnerable to environmental disturbance. Gyro integration was smooth but drifted, and the complementary filter was the most usable yaw estimate among the IMU-based options. GPS provided the best global trajectory, while accelerometer-only velocity and IMU dead reckoning accumulated large errors. LiDAR odometry produced a much more realistic local motion estimate than IMU-only integration, and camera data helped interpret route events and environmental causes of sensor degradation.

The overall conclusion is that robust navigation requires sensor fusion. IMU data should provide high-rate motion propagation, GPS should anchor the global frame when available, and LiDAR should provide local geometric correction in GPS-challenging environments. Camera data can further support interpretation or be extended into visual odometry in future work.